

13 (a) $\lambda = \ln 2 / T_{1/2}$
 $= \ln 2 / (53.3 \times 24 \times 60 \times 60) = 1.5 \times 10^{-7} \text{ s}^{-1}$
A1 [1]

(b) $A = \lambda N$ C1
 $N = 39 \times 10^{-3} / 1.5 \times 10^{-7} = 2.6 \times 10^5$
 $m = (2.6 \times 10^5 / 6.0 \times 10^{23}) \times 7 \times 10^{-3} \text{ or } 2.6 \times 10^5 \times 1.66 \times 10^{-27} \times 7$ C1
 $= 3.0 \times 10^{-21} \text{ kg}$ A1 [3]

(c) $2/39 = \exp(-1.5 \times 10^{-7} \times t) \text{ or } 2/39 = (1/2)^{[t/(53.3 \times 24 \times 3600)]}$ C1
 $t = 2.0 \times 10^7 \text{ s}$ A1 [2]